

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

TypeSense: An Adaptive Typing Tutor Based on Typing Behavior and Finger Analysis

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>Clockers</i> <i>DSCPP-III-2026-T217</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Typing tools usually only care about how fast you type and how many characters are right—so they miss all the little hiccups that actually slow you down, like wrong finger placement, awkward key jumps, hesitation or repeated corrections.

Because of this, people keep doing generic drills even when a specific part of their typing needs work. Our solution is to record every keystroke in detail: timestamps, error type, how long it takes between keys, and the pattern of any corrections. We'll also add a small computer-vision module that watches your fingers with a webcam and tells us which key each finger lands on.

By fusing those behavioural data points with the visual finger-tracking we build a learner profile that shows exactly where you're weak and how you're improving. The system then hands you practice exercises tailored to those gaps instead of throwing the same drills at everyone.

We're not reinventing the typing tutor; we're simply combining physical and behavioural cues into one adaptive engine written in C/C++ so learners get more focused feedback and faster progress.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Research has addressed several aspects of typing improvement. Soga et al. used webcams with coloured fingernail markers to spot fingering errors; TTracker employed Leap Motion for real-time finger tracking during touch-typing; Hagiya focused on statistical detection of input stumbles and personalized feedback; recent work uses Q-Learning to tailor exercises.

These studies show that finger placement, input difficulties, and adaptive content can be tackled separately. Our goal is to merge these signals into a single learner model, and we plan to test a low-cost approach using just a standard webcam with C++ rather than specialized hardware.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

- Build a typing tutor which does not just focus on speed and accuracy but analyses typing behaviour such as hesitation, corrections, repeated errors, and difficult key transitions.*
- Construct a learner profile by combining these behavioural patterns with details about finger use to identify more specific typing weaknesses.*
- Offer exercises tailored to the learner's real weaknesses instead of depending solely on general typing drills.*
- Find out if including finger-level information with keyboard data gives better personalization than relying solely on keyboard performance.*
- Determine if this method can reduce repeated errors and enhance typing performance and retention compared to fixed practice.*

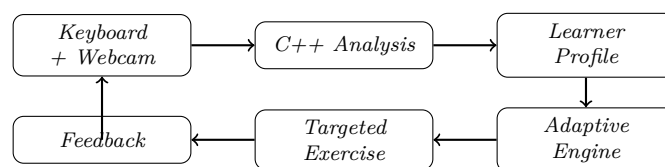
Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

- From each session the system will gather detailed information on typing, such as typing speed, accuracy, timing, errors, hesitations, corrections, and difficulties when switching between keys. This data will then be analysed in order to detect weak keys, repeated mistakes, difficult transitions, and correction habits.
- The results will be employed in order to establish a learner profile which monitors the user's strengths and weaknesses. The adaptive component will make use of this profile to choose exercises that address the learner's particular difficulties. We shall begin with a rule-based method and, if it is possible, look into Q-Learning as a means of selecting exercises.
- To carry out the physical analysis, we will use OpenCV to test a standard webcam in order to track finger movements and link finger usage to key presses and typing errors. If reliable tracking cannot be achieved, then this aspect of the study will remain optional.
- The system will include a native graphical user interface and will use SQLite to store the learners' profiles, session history, and performance data.
- We will then compare adaptive practice with fixed drills by assessing typing speed, accuracy, the number of repeated errors, hesitation, and retention in order to see if the individualised method has any effect on learning.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Interfaces: keyboard input enters the C++ core; webcam frames are handled by optional C++/OpenCV analysis; learner data is persisted in SQLite; the native Qt interface displays exercises, feedback and progress.

Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

Project Outcome

- *The key output will be a functional typing tutor featuring its own graphical user interface, a learner profile, a record of previous sessions, and exercises tailored to each individual user.*
- *The basic system will provide information on words per minute, accuracy, errors, hesitations, corrections, and repeated weak areas.*
- *We will also investigate whether incorporating finger-level data enhances personalization. The finger tracker will be included if it functions reliably; if not, the keyboard-based system will stay as the baseline.*
- *Lastly, we will compare adaptive practice with fixed drills by examining improvements in typing speed, reductions in errors, and the extent to which skills are retained.*
- *The project will also show whether combining data obtained from the keyboard with finger-level information can lead to more personalized typing practice.*

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

The system targets desktop/laptop touch-typing with a standard keyboard. A webcam is assumed only for the optional physical-analysis module. The first prototype focuses on English typing and a controlled exercise set. Finger tracking is experimental, so the baseline does not depend on perfect webcam detection. Main application logic remains in C/C++; GUI and database are supporting layers.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

- M. Soga, T. Tamura, and H. Taki, "Development of a Typing Skill Learning Environment with Diagnosis and Advice on Fingering Errors," *Procedia Computer Science*, vol. 22, pp. 737–744, 2013, doi: 10.1016/j.procs.2013.09.155.
- E. Kollie, F. Loizides, T. Hartley, and A. Worrallo, "TTracker: Using Finger Detection to Improve Touch Typing Training," in *Human-Computer Interaction – INTERACT 2017, Lecture Notes in Computer Science*, vol. 10516, pp. 469–472, Springer, 2017, doi: 10.1007/978-3-319-68059-0_53.
- T. Hagiya, T. Horiuchi, T. Yazaki, and T. Kawahara, "Typing Tutor: Individualized Tutoring in Text Entry for Older Adults Based on Statistical Input Stumble Detection," *Journal of Information Processing*, vol. 26, pp. 362–374, 2018, doi: 10.2197/ipsjip.26.362.
- G. Chrisnawati, Y. R. Sanjaya, A. Utomo, and N. Nurmiati, "Development of an Artificial Intelligence-Based Adaptive Typing Training System to Improve Accuracy and Speed," *Journal of Artificial Intelligence and Engineering Applications*, vol. 5, no. 2, 2026.
- W. K. Supriyatmoko and K. Anam, "Online Journal System," [Journal Name], vol. 5, no. 1, pp. 36–45, 2025.
- V. R. S. Reddy, K. Y. S. Preethi, T. Mounika, and N. Kousar, "AI Based IT Training System," *International Journal of Research Publication and Reviews*, vol. 6, no. 4, pp. 11291–11294, 2025.